

Technology Stack

Maximum Marks: 2 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the Rising Waters flood prediction solution. A well-chosen technology stack ensures the application is scalable, maintainable, and aligned with project requirements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Machine Learning	XGBoost 1.7 + Scikit-learn 1.3	Best accuracy (96.55%) among 4 tested algorithms; fast inference for real-time prediction
2	Data Processing	Pandas 2.x + NumPy 1.x	Efficient tabular data manipulation and array operations for the 5,000-row flood dataset
3	Model Serialization	Joblib 1.3	Save and load floods.save (XGBoost model) and scaler.save (StandardScaler)
4	Backend / Server-Side	Python 3.11 + Flask 2.x	Lightweight WSGI framework; rapid API development; integrates directly with ML objects
5	Frontend / Client-Side	HTML5 + CSS3 + Vanilla JS	Dark-theme glassmorphism UI; responsive design; no framework overhead
6	Templating Engine	Jinja2 3.x	Server-side HTML rendering; template inheritance via base.html eliminates duplication
7	Cloud / Hosting	IBM Cloud Foundry	Free-tier cloud hosting; seamless deployment via manifest.yml; IBM Skill Wallet alignment

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
8	Containerization	Docker 24+	Reproducible deployment environment; easy scaling; Dockerfile provided in repository
9	Version Control & CI/CD	Git + GitHub	Source code management; collaborative development; structured 8-phase documentation repository